SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: ***Apply image enhancement and segmentation techniques for extracting meaningful regions from images.*** **(BTL 3)**

Assessment Tool Weightage: 10%

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

Criteria	*Weightage*	*Level 4 – Exemplary*	*Level 3 – Proficient*	*Level 2 – Developing*	*Level 1 – Beginning*	*Marks(100)*
Understanding of Concepts	*25%*	*Demonstrates thorough, accurate understanding of concepts*	*Good understanding with minor gaps*	*Basic understanding with some misconceptions*	*Poor understanding; major misconceptions*	
Explanation	*25%*	*Detailed, well-explained answers with relevant examples*	*Clear explanation with adequate details*	*Limited explanation; lacks depth*	*Poor or unclear explanation*	
Application	*20%*	*Effectively applies concepts with strong analytical ability*	*Applies concepts with minor errors*	*Limited application with weak analysis*	*No application or incorrect analysis*	
Organization	*15%*	*Well-structured answers with logical flow and coherence*	*Organized with minor issues in flow*	*Basic structure, lacks clarity*	*Poor organization, incoherent answers*	

Accuracy	***10%***	***Completely accurate and covers all required points***	***Mostly accurate with minor omissions***	***Partially correct with missing points***	***Incorrect answers with major gaps***	
Clarity	***5%***	***Clear, neat, professional presentation***	***Understandable with minor issues***	***Limited clarity, readability issues***	***Poorly written, difficult to understand***	

1.An image appears too bright.
Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

2.A dark image lacks detail.
Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

3.An image has poor contrast.
Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

4A low-contrast image requires uniform intensity distribution.
Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

5.An image shows uneven brightness across regions.
Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

6.A noisy image contains random intensity variations.
Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

7.An image requires noise reduction without significant loss of details.
Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

8.An image appears blurred after smoothing.
Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

9.Fine details in an image are not clearly visible.
Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

10A noisy image must be preprocessed before further analysis.
Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.